Lecture 11

Pete Manolios
Northeastern

Logistics

- Post on Piazza re. exam
- Post on Papers & Projects
  - Speak to me if you haven't done so already

First Order Logic

- Example: Group Theory
  - (G1) For all x, y, z: $(x \cdot y) \cdot z = x \cdot (y \cdot z)$
  - (G2) For all x: $x \cdot e = x$
  - (G3) For all x there is a y such that: $x \cdot y = e$
- Theorem: For every x, there is a y such that $y \cdot x = e$
- Examples of groups: Nat, +, 0?; Int, +, 0?, Real, *, 1?
- Proof:

  By (G3) there is: a y s.t. $x \cdot y = e$ and a z s.t. $y \cdot z = e$

  Now: $y \cdot x = y \cdot x \cdot e = y \cdot x \cdot y \cdot z = y \cdot e \cdot z = y \cdot z = e$
- Is this true for all groups? Why?
- How many groups are there?
- Are there true statements about groups with no proof?

First Order Logic

- First Order Logic forms the foundation of mathematics
- We study various objects, e.g., groups
- Properties of objects captured by “non-logical” axioms
  - (G1-G3 in our example)
- Theory consists of all consequences of “non-logical” axioms
  - Derivable via logical reasoning alone
  - That’s it; no appeals to intuition
- Separation into non-logical axioms logical reasoning is astonishing: all theories use exactly same reasoning
- But, what is a proof ($\Phi \vdash \phi$)?
- Question leads to computer science
- Proof should be so clear, even a machine can check it

First Order Logic: Syntax

- Every FOL (first order language) includes
  - Variables v_0, v_1, v_2, ...
  - Boolean connectives: $\vee$, $\neg$
  - Equality: =
  - Parenthesis: (,)
  - Quantifiers: $\exists$
- The symbol set of a FOL contains (possibly empty) sets of
  - relation symbols, each with an arity > 0
  - function symbols, each with an arity > 0
  - constant symbols
- Example: groups 2-ary function symbol • and constant e
- Set theory: $\in$, a 2-ary relation symbol, ...

First Order Logic: Terms

- Terms denote objects of study, e.g., group elements
- The set of S-terms is the least set closed under:
  - Every variable is a term
  - Every constant is a term
  - If $t_1, \ldots, t_n$ are terms and f is an n-ary function symbol, then $f(t_1, \ldots, t_n)$ is a term

First Order Logic: Formulas

- Formulas: statements about the objects of study
- An atomic formula of S is
  - $t_1 = t_2$ or
  - $R(t_1, ..., t_n)$, where t_i is an S-term and R is an n-ary relation symbol in S
- The set of S-formulas is the least set closed under:
  - Every atomic formula is a formula
  - If ϕ, ψ are S-formulas and x is a variable, then $\neg\phi$, $(\phi \vee \psi)$, and $\exists x\phi$ are S-formulas
- All Boolean connectives can be defined in terms of $\neg$ and $\vee$
- We can define $\forall x\phi$ to be $\neg\exists x\neg\phi$

Definitions on Terms & Formulas

- Define the notion of a free variable for an S-formula
- The definition of formula depends on that of term
- So, we're going to need an auxiliary definition:

  $var(x) = \{x\}$

  $var(c) = \{\}$

  $var(f(t_1, ..., t_n)) = var(t_1) \cup \cdots \cup var(t_n)$
- Is this a definition? (termination!)

  $free(t_1 = t_2) = var(t_1) \cup var(t_2)$

  $free(R(t_1, ..., t_n)) = var(t_1) \cup \cdots \cup var(t_n)$

  $free(\neg\phi) = free(\phi)$

  $free((\phi \vee \psi)) = free(\phi) \cup free(\psi)$

  $free(\exists x\phi) = free(\phi) \setminus \{x\}$

Semantics of First Order Logic

- What does $\exists v_0 R(v_0, v_1)$ mean?
- It depends on:
  - What R means (what relation over what domain?)
  - What v_1 means (what element of the domain?)
- What if the is domain $\mathbb{N}$, R is $<$, and v_1 is 1? If v_1 is 0?
- An S-interpretation $\mathcal{I} = \langle A, a, \beta \rangle$ where ($\langle A, a \rangle$ is an S-structure)
  - A is a non-empty set (domain or universe)
  - a is a function with domain S
  - β: Var $\rightarrow A$ is an assignment
  - If $c \in S$ is a constant, then $a.c \in A$
  - If $f \in S$ is an n-ary function symbol, then $a.f : A^n \rightarrow A$
  - If $R \in S$ is an n-ary relation symbol, then $a.R \subseteq A^n$

Meaning via Interpretations

- The meaning of a term in an interpretation $\mathcal{I} = \langle A, a, \beta \rangle$
  - If $v \in Var$, then $\mathcal{I}.v = \beta.v$
  - If $c \in S$ is a constant, then $\mathcal{I}.c = a.c$
  - If $f(t_1, ..., t_n)$ is a term, then $\mathcal{I}(f(t_1, ..., t_n))$ is $(a.f)(\mathcal{I}.t_1, ..., \mathcal{I}.t_n)$
- What it means for an interpretation to satisfy a formula:
  - $\mathcal{I} \models (t_1 = t_2)$ iff $\mathcal{I}.t_1 = \mathcal{I}.t_2$
  - $\mathcal{I} \models R(t_1, ..., t_n)$ iff $\langle \mathcal{I}.t_1, ..., \mathcal{I}.t_n \rangle \in a.R$
  - $\mathcal{I} \models \neg\phi$ iff not $\mathcal{I} \models \phi$
  - $\mathcal{I} \models (\phi \vee \psi)$ iff $\mathcal{I} \models \phi$ or $\mathcal{I} \models \psi$
  - $\mathcal{I} \models \exists x\phi$ iff for some $b \in A$, $\mathcal{I}(x \leftarrow b) \models \phi$

Models & Consequence

- Let Φ be a set of formulas and ϕ a formula
- $\mathcal{I} \models \Phi$ ($\mathcal{I}$ is a model of Φ) iff for every $\phi \in \Phi$, $\mathcal{I} \models \phi$
- $\Phi \models \phi$ (ϕ is a consequence of Φ) iff for every interpretation, $\mathcal{I}$, which is a model of Φ, we have that $\mathcal{I} \models \phi$
- ϕ is *valid* iff $\varnothing \models \phi$, which we write as $\models \phi$
- A formula ϕ is satisfiable, written *Sat* ϕ, iff there is an interpretation which is a model of ϕ
- A set of formulas Φ is satisfiable (*Sat* Φ), iff there is an interpretation which is a model of all the formulas in Φ

SAT & Validity

- Lemma: For all ϕ, Φ: $\Phi \models \phi$ iff not *Sat* $(\Phi \cup \{\neg\phi\})$
- Proof $\Phi \models \phi$

  iff for all $\mathcal{I}$, $\mathcal{I} \models \Phi$ implies $\mathcal{I} \models \phi$

  iff there is no $\mathcal{I}$ such that $\mathcal{I} \models \Phi$ but not $\mathcal{I} \models \phi$

  iff there is no $\mathcal{I}$ such that $\mathcal{I} \models \Phi \cup \{\neg\phi\}$

  iff not *Sat* $\Phi \cup \{\neg\phi\}$
- As a consequence, ϕ is valid iff $\neg\phi$ is not satisfiable

Examples

- Consider symbol sets $S_{ar} := \{+,\cdot,0,1\}$ and $S_{ar}^{<} := \{+,\cdot,0,1,<\}$
- N denotes the S_{ar}-structure $\langle \omega,+^{\omega},\cdot^{\omega},0^{\omega},1^{\omega}\rangle$, where $+^{\omega},\cdot^{\omega},0^{\omega},1^{\omega}$ correspond to $+,\cdot,0,1$ on ω
- $N^{<}$ denotes the $S_{ar}^{<}$-structure $\langle \omega,+^{\omega},\cdot^{\omega},0^{\omega},1^{\omega},<^{\omega}\rangle$, where $<^{\omega}$ corresponds to $<$ on ω
- R denotes the S_{ar}-structure $\langle R,+^{R},\cdot^{R},0^{R},1^{R}\rangle$, where R is the set of real numbers
- $R^{<}$ denotes the $S_{ar}^{<}$-structure $\langle R,+^{R},\cdot^{R},0^{R},1^{R}, <^{R}\rangle$, where $+^{R},\cdot^{R},0^{R},1^{R}$, $<^{R}$ correspond to $+,\cdot,0,1,<$ on R
- $+^{R}$ and $+^{\omega}$ are very different objects, but we will drop the subscripts when (we think) no ambiguity will arise

Coincidence Lemma

- Let $\mathcal{I}_1 = \langle A, a_1, \beta_1 \rangle$ be an S_1-interpretation and let $\mathcal{I}_2 = \langle A, a_2, \beta_2 \rangle$ be an S_2-interpretation (both have the same domain). Let $S = S_1 \cap S_2$.
  - 1. Let t be an S-term. If $\mathcal{I}_1$ and $\mathcal{I}_2$ agree on the S-symbols occurring in t and on the variables occurring in t, then $\mathcal{I}_1(t) = \mathcal{I}_2(t)$.
  - 2. Let ϕ be an S-formula. If $\mathcal{I}_1$ and $\mathcal{I}_2$ agree on the S-symbols and on the variables occurring free in ϕ, then $\mathcal{I}_1 \models \phi$ iff $\mathcal{I}_2 \models \phi$.
- Proof: By induction on S-terms and then on S-formulas
- This is a very useful lemma

Substitution

- Substituting *t* for *x* in ϕ yields ϕ', which says about *t* what ϕ says about *x*
- Consider ϕ = ∃*z* *z*+*z* ≡ *x*. Note that ⟨*N*,*β*⟩ ⊨ ϕ iff *β.x* is even
  - Replacing *x* by *y* gives, ϕ′ = ∃*z*z+*z*≡*y* ,where ⟨*N*,*β*⟩ ⊨ ϕ' iff *β.y* is even; good!
  - What about replacing *x* by *z*? This gives ϕ′ =∃*z*z+*z*≡*z*, but N ⊨ ϕ'; bad!
  - Have to deal with variable capture
    - The book provides a definition which replaces bound occurrences of *z* with a new variable in ϕ
- Theorem: For every term, *t*, $\mathcal{J}(t\frac{t_0\ldots t_r}{x_0\ldots x_r}) = \mathcal{J}\frac{\mathcal{J}(t_0)\ldots\mathcal{J}(t_r)}{x_0\ldots x_r}(t)$
- Theorem: For every formula, ϕ, $\mathcal{J} \vDash \phi\frac{t_0\ldots t_r}{x_0\ldots x_r}$ **iff** $\mathcal{J}\frac{\mathcal{J}(t_0)\ldots\mathcal{J}(t_r)}{x_0\ldots x_r} \vDash \phi$
- Theorem: If ϕ is Valid then so is $\phi\frac{t_0\ldots t_r}{x_0\ldots x_r}$

Formalization Examples

$\forall x Rxx$

$\forall x \forall y (Rxy) \Rightarrow (Ryx)$

$\forall x \forall y \forall z ((Rxy \wedge Ryz) \Rightarrow Rxz)$

Equivalence relations

$\langle \forall x :: xRx \rangle$

$\langle \forall x, y :: xRy \Rightarrow yRx \rangle$

$\langle \forall x, y, z :: xRy \wedge yRz \quad \Rightarrow \quad xRz \rangle$

The way I would write it

Define a new quantifier "there exists exactly one," written $\exists^{=1} x\phi$

Try it!

$\exists x (\phi \wedge \forall y (\phi \frac{y}{x} \Rightarrow x = y))$

Proof Theory

- $\Phi \vdash \phi$ denotes that ϕ is provable from Φ
- Provability should be machine checkable
- It may seem hopeless to nail down what a proof is
  - don't mathematicians expand their proof methods?
- FOL has a fairly simply set of obvious rules
- There are many equivalent ways of defining proof